

THE USER



Opens Verita



Drops a CSV



Reads dashboard



Asks in English



Runs Investigator

INTERFACE

Landing.tsx

Studio dropzone

ECharts canvas

NL → SQL console

Investigator panel

each action fires the pipeline below

PIPELINE

FastAPI router

Pandas profiling

scipy insights

DuckDB (sandbox)

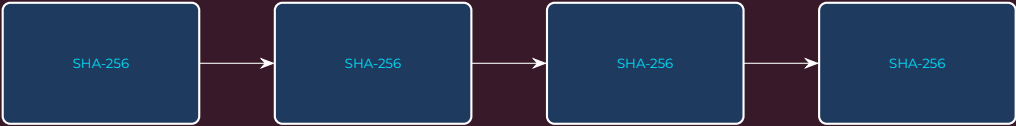
AI Agent + Gemini

SHA-256 sealed

THE AGENT TESTS 5 REAL AML TYPOLOGIES

- 1 Geographic risk concentration
 - 2 Channel risk concentration
 - 3 Structuring near the \$10,000 threshold
 - 4 Large-value concentration (top-1% share)
 - 5 Temporal activity spike (mean + 2σ)
- each tested with a real **DuckDB** query + statistic.

EACH STEP IS HASH-CHAINED



commits id · query · finding · severity · timestamp · prev-hash
doctor any step → the chain breaks.

PIPELINE FACTS

- ✓ 6 REST routers
- ✓ 3 service layers
- ✓ DuckDB sandboxed
- ✓ TreeSHAP (exact)
- ✓ ROC-AUC 0.913
- ✓ PR-AUC 0.65
- ✓ SHA-256 verified
- ✓ Gemini optional
- ✓ rule-based fallback

Every stage logged; every number computed at request time — no hardcoded metrics.